System Overview:

DBS Planning & Validation Workstation with 6 modules:

1. Patient Loader (DICOM → volume + metadata)
2. Multi-planar MRI Viewer (MPR + arbitrary slicing)
3. AI Module (STN/RN segmentation + landmarks)
4. Registration Module (T1 ↔ T2 ↔ CT)
5. Planning Module (trajectory definition + validation)
6. Post-op Module (electrode detection + evaluation)
7. 3D Reconstruction (shared across modules)

Modules:

Patient Loader:

This module is responsible for importing and organizing patient data.
It allows the user to:

- Load a full DICOM folder containing MRI or CT scans
- Alternatively, load preprocessed NIfTI files
  Once loaded, the software:
- Reconstructs the scans into a 3D volume
- Extracts essential information such as image orientation, voxel size, and patient metadata
  The output is a standardized internal representation of the patient's imaging data that can be used consistently across all modules

MRI Viewer (Visualization Module):

This module provides an interactive interface for exploring medical images.
The user can:

- View MRI scans in three standard orientations:
  - Axial
  - Sagittal
  - Coronal
- Navigate through slices using scrolling
- Adjust image contrast (window/level)
- Visualize overlays such as:

    - Segmentation masks (STN, RN)
    - Anatomical landmarks
    - Planned trajectories
      Additionally, the viewer supports synchronized navigation between views and may allow arbitrary angled slicing for advanced inspection.

AI Module (Segmentation and Anatomical Landmarks):

This module performs automated analysis of MRI data.
It:

- Segments key brain structures:
    - Subthalamic Nucleus (STN)
    - Red Nucleus (RN)
      From these segmentations, the system defines the **Bejjani line** that serves as an anatomical reference
      The module also supports defining the **AC–PC line**, either:
- Manually (by user selection), or
- Semi-automatically (optional advanced feature)
- then predict optimal coordinates based on AC PC reference
  All relevant anatomical points are then expressed in a consistent coordinate system for planning

Registration Module:

This module aligns images from different modalities into the same spatial reference.
It enables:

- Alignment of T1 and T2 MRI scans
- Alignment of CT scans with MRI
  The goal is to ensure that all data (anatomy, segmentation, electrodes) are spatially consistent.
  This is essential for:
- Accurate planning
- Reliable post-operative analysis

Planning Module:

This module supports surgical trajectory planning.
The user can:

- Select a target point (typically within the STN)

- Define an entry point on the skull
  The software then:
- Generates a trajectory between the two points
- Displays this trajectory across all image views
  It also allows the user to:
- Inspect the path slice by slice
- Verify that the trajectory avoids critical structures
  This provides a visual and geometric validation of the surgical plan.

3D Reconstruction Module

This module generates 3D representations of anatomical structures and planning elements.
It allows visualization of:

- STN and RN as 3D objects
- The planned electrode trajectory
- (Optionally) the brain volume
  This helps the user better understand spatial relationships and improves planning accuracy

Post-operative Module (CT Analysis)

This module evaluates electrode placement after surgery.
Using CT scans, the software:

- Detects electrodes based on their high intensity
- Identifies their positions automatically
  After aligning the CT with pre-operative MRI, the system:
- Displays electrode locations relative to the STN
- Allows both 2D and 3D visualization

Evaluation and Quantification

The software provides quantitative measures such as:

- Distance between electrode and target (STN)
- Degree of overlap between electrode and STN
  These metrics help assess the accuracy of the surgical outcome.

Licensing System (Optional Feature)

The software may include a licensing mechanism to restrict usage.
It ensures that:

- The software runs only on authorized machines
- Access is controlled through a unique activation key

Deployment and Technology Stack:

The proposed system will be developed as a **desktop application running on Windows**. This choice ensures efficient handling of large medical imaging data (MRI and CT scans) and provides a responsive and interactive user experience required for clinical workflows.
The software will be implemented using the **Python programming language**, leveraging its strong ecosystem of scientific and medical imaging libraries.
Key technologies include:

- Python for core application development
- PyQt for designing the graphical user interface
- SimpleITK for image processing and registration
- nibabel for handling medical image formats
- PyTorch (with MONAI) for implementing the segmentation models
  This stack enables the development of a robust, modular, and scalable application tailored to medical image analysis and neurosurgical planning.